



Aryan Kumar
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22B2423
B.Tech.
Gender: Male
DOB: 03/10/2004

Examination	University	Institute	Year	CPI / %
Graduation	IIT Bombay	IIT Bombay	2026	8.29

Pursuing a **Minor Degree** in **Artificial Intelligence** and **Data Science** (Minor CPI: 9) IIT Bombay | **CFA Level 1** Candidate (Aug '25), USA

PROFESSIONAL EXPERIENCE

MOBAVENUE MEDIA Data Science Intern [Feb '25 - Aug '25]	<i>Recognized with LOR by Co-founder; Ad-tech firm specializing in data-driven marketing and programmatic advertising</i> - Achieved 85% recall for installs, 70% for purchases with TabNet across 10B+ of impressions boosting ROAS - Identified 8K+ top inventories from 58K+, enabling targeted and cost-efficient ad bid placement with <4ms latency - Reduced bidding costs by 18% by redesigning formula with user scores, inventory tiers and hourly pacing trends
PHILLIPCAPITAL INDIA Strategy Intern [May '25 - Jul '25]	<i>Received a Letter of Recommendation from CEO in recognition of exceptional contributions and high-impact results</i> - Studied 15+ firms; identified 20+ gaps, proposed solutions to VP targeting \$10-12B market potential by FY2030 - Targeted 500K+ users by proposing P9 Lite discount brokerage & global investing without cannibalizing P9 users - Secured C-Suite approval through board presentations, leading to launch and high-volume revenue streams
CUSTOMERSHASTRA ML Research Intern [May '24 - Jul '24]	<i>Boosted prioritization efficiency by 80% with spaCy-EN model to score product feedback, accelerating decision-making</i> - Reduced review cycles from 720 hrs to seconds across 10K+ feedback cases by creating 40+ high-impact NLP feature - Automated customer-agent email audits, enabled quality checks on 2K+ emails/month and 46% boost in compliance

RESEARCH & INTERNATIONAL EXPOSURE

TOYOTA TECHNOLOGICAL INSTITUTE, JAPAN Research Intern [May '24 - Jul '24]	<i>Received LOR for supercapacitor/CNT R&D in a 3-month, 5+ country, 30+ researcher lab under Prof. Masamichi Yoshimura</i>
Approach	- Developed electrode fabrication after reviewing 20+ academic papers on supercapacitor performance - Automated data processing using 10+ python libraries boosting efficiency by removing redundant tasks - Designed nano-petal Ni-Foam/Co/Mo/Ni electrodes with controlled synthesis time and concentration
Result	- Achieved a specific capacitance of 837 F/g (4.65x industry standard) influencing future R&D planning - Reduced data handling time by 90%, enabling faster experiment turnaround and decision-making - Doubled deposition density and produced 5 µm nano-petals by extending synthesis from 8 hrs to 24 hrs

TECHNICAL PROJECTS

SATELLITE IMAGE SEGMENTATION AND CHANGE DETECTION ISRO Smart India Hackathon <i>Ranked 5th nationally among 200+ finalists, one of only 3 teams from IIT Bombay, recognized for innovative solution by MoE, India</i> - Achieved 92% accuracy on 20K+ images; led a team of 6 to develop a 41-layer U-Net CNN using pixel-wise semantic segmentation - Conducted comprehensive literature review of 10+ geospatial models to benchmark and identify the optimal architecture for satellite images - Proposed potential monitoring of 1100+ border villages by integrating the model into a scalable change-detection framework recognized by ISRO	[Nov '23-Dec '23]
ALGORITHM VISUALIZER Learner's Space, IIT Bombay <i>Engineered a neural network from scratch for high-accuracy image-based handwritten digit and text recognition with real-time GUI Integration</i> - Achieved 96% accuracy over 15 epochs by building a 3-layer deep neural network using NumPy to classify EMNIST handwritten characters - Optimized training using stochastic gradient descent with mini batches of 128 samples, accelerating the model convergence by 25% - Developed an OpenCV-based interactive drawing GUI enabling real-time predictions, used by 20+ users during tests to validate the model	[Jun '25-Jul '25]
HANDWRITTEN CHARACTER RECOGNITION USING NEURAL NETS WiDS Analytics Club, IIT Bombay <i>Engineered a high-frequency, real-time bid-ask order scraper with GUI using PyTesseract, processing 10+ screen regions with <2ms latency</i> - Achieved 96% accuracy over 15 epochs by building a 3-layer deep neural network using NumPy to classify EMNIST handwritten characters - Optimized training using stochastic gradient descent with mini batches of 128 samples, accelerating the model convergence by 25% - Developed an OpenCV-based interactive drawing GUI enabling real-time predictions, used by 20+ users during tests to validate the model	[Dec '23-Jan '24]
REAL-TIME ORDER BOOK SCRAPING & AUTOMATED TRADING Freelance <i>Engineered a high-frequency, real-time bid-ask order scraper with GUI using PyTesseract, processing 10+ screen regions with <2ms latency</i> - Engineered a high-frequency, real-time bid-ask order scraper with GUI using PyTesseract, processing 10+ screen regions with <2ms latency - Developed & backtested 15+ trading strategies on large-scale order book data, evaluating performance across 100K+ simulated trades - Automated trade execution pipeline, delivering 4% higher returns on TATA Motors and reducing manual monitoring overhead by 95%	[Feb '25-Jul '25]
ARCHITECTURAL LAYOUT CLASSIFICATION Course Project: DS203 <i>Improved design retrieval speed by 30% by classifying 10000+ building layouts into 8 complexity clusters using GridSearchCV hyper tuning</i> - Engineered an image processing model with Canny edge detection and contour-based feature extraction to quantify layout complexity - Integrated classification model into a searchable database, enabling rapid retrieval and comparison of architectural layouts for designs	[Jan '24-Apr '24]
NEUROMAPPING AI for Early Neurodiversity Detection Course Project: DH307 <i>Building an AI/ML diagnostic tool to detect neurodiverse conditions affecting 15-20% of global population, targeting >90% accuracy</i> Designing analytics pipelines to process 10K+ patient records on dyslexia, ADHD, dyspraxia, and autism for model training and validation	[Jul '25- Present]

REAL-TIME SOCIAL DISTANCING MONITORING SYSTEM | Computer Vision Project | Self-Project

[Sep '22-Nov '22]

- Engineered a Python-based system using OpenCV, YOLO, and SciPy to detect and track people in real time for social distancing compliance
- Implemented the **Euclidean distance algorithm** to measure inter-person spacing with high accuracy, triggering visual alerts for violations
- Integrated video I/O pipelines with live overlays and auto-recording of **flagged** events leads to reducing manual surveillance effort by **40%**

TEXT-TO-SPEECH CONVERSION USING GAUSSIAN DIFFUSION MODELS | Self-Project

[Oct '24- Nov '24]

- Developed transformer encoder with windowed **self-attention** and **feedforward** layers to improve phoneme-to-representation from inputs
- Used Probabilistic Diffusion Models to gradually enhance pure **noise** into target audio given input text, by solving a Time-Reversed **SDE**'s
- Validated audio quality through **spectrogram** similarity and listening tests, showing noticeable improvement over baseline **TTS** models

SINGLE & MULTI-LEVEL THRESHOLDING OF IMAGES USING KITTLER-ILLINGWORTH ALGORITHM | Course Project

[Jan '25-Apr '25]

- Accurately segmented images using Poisson-based Kittler-Illingworth thresholding, achieving **0.86** correlation and **0.82** JS Divergence
- Integrated **GMM** into Kittler-Illingworth framework for multi-thresholding, computing **4** optimal thresholds to classify pixels into **5** regions
- Validated algorithm robustness on noisy, non-Gaussian datasets using good region uniformity (**2113.7**) and second-order entropy (**1.338**)

FINANCE PROJECTS

CORPORATE FINANCE | Finance Club, IIT Bombay

[Jun '25-Aug '25]

- Built a detailed **DCF** valuation model for sugar sector firms, incorporating WACC, M&A scenarios, and sensitivity analyses
- Optimized capital structure, reducing WACC by **2%** and driving a projected **10%** improvement in **ROI** through **debt-equity** rebalancing
- Leveraged **NPV** and **IRR** analysis to optimize investment decisions, resulting in **15%** higher liquidity and **20%** improved profitability

FINANCIAL MODELING | Finance Club, IIT Bombay

[Jan '24-Feb '24]

- Performed discounted cash flow (**DCF**) models to evaluate receivables and payables for a firm's cash flows and assess financial health
- Mitigated interest rate risk via structured swap agreement, hedging against adverse market movements and reducing volatility exposure
- Developed a swap pricer using no-arbitrage and bootstrapping principles, achieving **<0.5%** valuation error for precise risk management

SUMMER OF QUANT | Quant Community | Institute technical Council | IIT Bombay

[May '25- Jul '25]

- Completed **8-weeks** intensive Quantitative Finance bootcamp covering Time Series Analysis, Probability, and Financial Derivatives
- Developed and trained LSTM-based stock forecasting model, achieving **92%** directional accuracy and **1.85** RMSE on historical data

OPTIONS PRICING | Fin-Search, Finance Club, IIT Bombay

[Jun '24-Aug '24]

- Achieved **70.18%** accuracy on Bank Nifty Call Options (**14-day** expiry) using Python-based Black–Scholes models integrating **4** option greeks
- Attained **81%** accuracy with a sequential Artificial Neural Network (**ANN**) leveraging option chain data to enhance pricing insights for trader

POSITIONS OF RESPONSIBILITY

MANAGER Innovation Council IIT Bombay MoE [Oct '24 – May '25]	<i>Strengthening IITB's entrepreneurship and innovation ecosystem to drive toward top NIRF ranking by 2026</i>			
	- Initiating 5+ nationwide innovation competitions and developing a newsletter to showcase IITB's achievements - Procured INR 2M+ grants & spearheaded pitches to 8 corporate leaders, fostering international partnerships - Amplified social media engagement and website traffic by 40% across 3 platforms, reaching 10K+ global audience			
DEPARTMENT ALUMNI SECRETARY Student Alumni Relations Cell [Jun '23 – Mar '24]	<i>Represented MEMS as part of a 15-member team fostering relations between 60K+ Alumni and 12K+ students</i>			
	- Procured 80+ student internships by contacting IITB alumni under ILP, improved their professional experience - Connected 650+ students and 300+ mentors through Placement Mentoring Program, SARC for career support - Executed SARCathon competitions at PAN-IIT level, engaging over 500+ participants with prizes worth 0.2M+			

TECHNICAL SKILLS

LANGUAGES	C, C++, LATEX, SQL, HTML, CSS, R Python (Matplotlib, Seaborn, Pandas, NumPy, SciPy, Scikit-learn, TensorFlow, Keras, PyTorch, XGBoost, LightGBM, Polars, OpenCV, Pillow, scikit-image, pytesseract , NLTK, spaCy, TextBlob, manim)
SOFTWARE	VSCode, Xcode, Docker, Linux, AWS (S3, Athena, Glue), GCP, NotebookLM, Colab, MS Office, Power BI, TradingView
KEY COURSES	Mathematical Optimization, Advanced Methods in Image Processing, Data Science, Machine Learning, Programming in C, R&D Project in AI/ML, Calculus, Differential Equations, Linear Algebra, Entrepreneurship Courses, Fundamentals of Digital Image Processing, Stop-Motion Animation, Molecular Simulations, Numerical Methods, Philosophy, Arts, Psychology

ACCOLADES & EXTRA-CURRICULARS

SCHOLASTIC	JEE Mains: 98.70 percentile in 1.1M+ students ['22] WBJEE: Secured AIR 773 out of 0.1M+ students ['22] - Secured top AA/AB grades across 21 courses ['22 - '25] - Selected in Magadh Super 30 out of 10K+ aspirants ['20]			
COMPETITIONS	Position	Competition	Significance	Details
	Institute Rank: 5th	ResCon, TechConnect	50+ Teams	Research Poster Presentation Competition
	Institute Rank: Top 15	Portfolio Management Competition	150+ Teams	Arranged by Finance Club, IIT Bombay
	Institute Rank: 4th	HULT Prize, IIT Bombay	60+ Teams	Organized by The Hult Prize Foundation
CERTIFICATIONS	- Completed 33 Hr DeepLearning.AI course, Stanford University Excel, SQL and Latex by Learner's space, IITB BCG Virtual Program Bloomberg Essentials 6 Hr JP Morgan Chase & Co Investment Banking by Forage			
MISCELLANEOUS	- Received a prestigious competitive Research Internship Award of ¥196,000 for Summer Internship in Japan - Taught 50+ underprivileged students in STEM by leading hands-on experiments and creating tailored question sets - Hosted SARC exam series and UPSC CSE 2020 AIR 1 & 110 panel discussion, engaging 200+ participants			
INTERESTS & HOBBIES	Trekking Mountains & Trails, Traveling, Cooking, Learning languages (German, Japanese), Geopolitics, Hollywood			